Task 12 - Adapt the Design for Smart Meters

C Programming Assignment - Rural Electricity Billing System

New requirements

Smart meters send watt-hour readings every 15 minutes, giving 35,040 readings per household per year. Tariffs can change at runtime and electricity used from 6 PM inclusive to 10 PM exclusive costs 20% more.

Data type impact

Use uint64_t for cumulative watt-hours and int64_t for money in paise. A 32-bit unsigned counter can store up to 4,294,967,295 Wh; a 64-bit value gives much more headroom. Check yearly sums for overflow.

Maximum values

An exact absolute maximum cannot be calculated because the prompt does not state maximum Wh per reading, maximum meter count or maximum households. If maximum interval energy is E Wh, annual energy is at most E × 35,040; use checked 64-bit arithmetic.

Peak pricing

For energy used from 18:00 inclusive to before 22:00, apply 120/100 of the normal rate. Split intervals crossing 18:00 or 22:00 so only peak-period energy receives the surcharge. Define rounding for fractional paise.

Runtime configuration

Load tariffs from a validated configuration file and store tariff version and effective time with each bill. Reject invalid rates and preserve audit information.

Pseudocode

For each interval: validate timestamp and Wh; split interval at tariff boundaries; select the effective tariff; apply 120/100 for peak segments; add each charge with overflow checks; record tariff version and audit data.

Trade-offs and tests

Wide integers improve range but need overflow checks. Runtime configuration improves flexibility but requires validation and security. Test 17:45, 18:00, 21:45 and 22:00; crossing intervals; maximum readings; 35,040 records; missing, duplicate or out-of-order records; tariff changes and overflow.

Important: illustrative tariff rates are used because the screenshot does not show the complete official tariff table. Replace them with your instructor's prescribed rates.